ASSIGNMENT 3 Coastal Engineering

Wave Field for Construction of a New Pier

ENCN442-26S1

Surface Water, Ground Water and Coastal Engineering

Zac Hemmings

zhe32@uclive.ac.nz

University of Canterbury

Date Submitted: 13 May 2026

Generative AI was used to assist with editing and formatting in this report in accordance with the ENCN442 course guidelines.

Assignment Duration about 2 Hours

1. UP-CROSSING ANALYSIS

After loading the 14,400 results at Δt = 0.25 s, I subtracted the mean to centre the signal on zero. Using np.sign(η) I identified all zero up crossings, then linearly interpolated crossing times between adjacent samples. Each pair of consecutive crossings defines one wave: height H = max(η) − min(η) over the interval, period T = Δt between crossings.

The record contains 275 waves:

Tz = 13.02 s

hs = 0.516 m - taken from mean of top 91

$H_{1/100}$ = 0.725 m - taken from mean of top 2

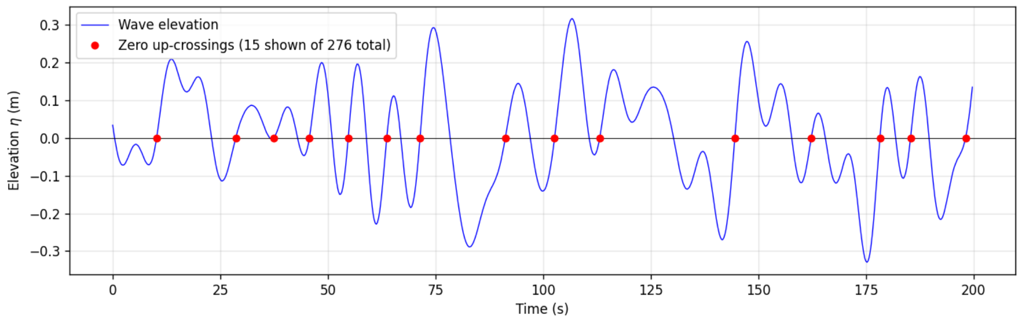

Figure 1: First 200s of wave crossings

2. WAVE CLASSIFICATION

I solved $\omega^2 = gk \cdot \tanh(kh)$ numerically using Tz at each depth calculated previously in Q1.

For measurement:

h = 17 m

λ_1 = 156.8 m

h/ λ = 0.108

Classification: This wave is Transitional.

For design:

h = 0.9 m

λ_2 = 38.5 m

h/ λ = 0.023

Classification: This wave is Shallow

Group velocities:

Cg1 = 10.52 m/s, cg_2 = 2.94 m/s

Using Snells law, With α_1 = 25° given at the measurement location, Snell's law $\sin \alpha_2 = (\lambda_2/\lambda_1) \sin \alpha_1$ gives:

α_2 = 5.96°, we can then calculate:

Coefficients: $Ks = \sqrt{(cg_1/cg_2)} = 1.891$

$Kr = \sqrt{(\cos \alpha_1/\cos \alpha_2)} = 0.955$

Design parameters:

$H_{1/100}$ = 0.725 x Ks x Kr = 1.3 m

Tz = 13 s (no change)

α = 5.96°

H/h = 1.45 at the design site is well above the breaking threshold of κ = 0.78, so the wave would break before reaching the pier, this suggests that the linear result is an idealisation of true wave characteristics for this case.

3. MORISON – EQUATION FORCES ON THE PIER

Using $H_{1/100}$ = 1.3 m from Q2 as the design wave height, I evaluated linear wave theory at z = 0. For shallow water we can assume the orbital velocity and acceleration are uniform irrespective of depth, so the choice of z has little effect on the result.

At z = 0 in shallow water I evaluated linear wave theory over one period:

$u(t) = (H\omega/2)\cdot\coth(kh)\cdot\cos(\omega t) \rightarrow u_{max} = 2.17$ m/s

$a_x(t) = (H\omega^2/2)\cdot\coth(kh)\cdot\sin(\omega t) \rightarrow a_{x,max} = 1.05$ m/s²

Then evaluated the Morrison equation with ρ = 1025 kg/m³, D = 0.3 m, Cd = 0.9, Ci = 2.

Peak F_D = 650 N/m

Peak F_I = 77 N/m

Max total force per unit length approx. 653 N/m

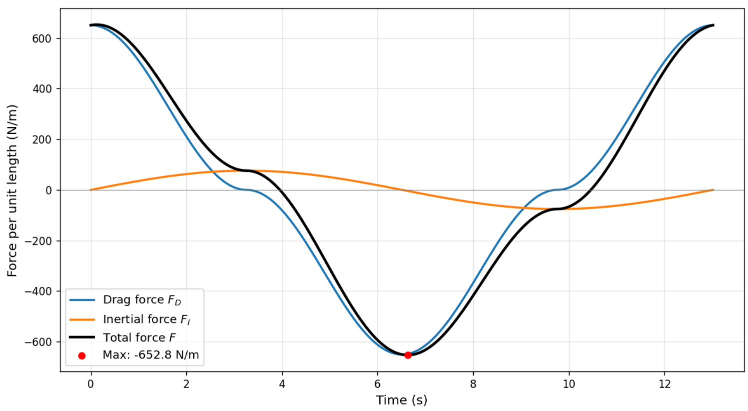

Figure 2: Morrison equation results, for analysis of force per unit length on pier support